

Lab Activity: Building a Digital Music Manager

CSCI 236 - Data Structures
Using Doubly Linked Lists

October 1, 2025

I. Objectives

To design and implement a dynamic doubly linked list in Python using classes to allocate and de-allocate nodes, and perform operations such as inserting, deleting, editing, traversing, and displaying items in the list.

II. Overview & Requirements

You are tasked with building a Digital Music Manager (DMM) in Python. The DMM must support a menu of commands to manage a playlist of songs using a doubly linked list. The playlist will be read from a CSV file and each song will be stored as a node in the list. You must implement the following classes:

SongRecord

- SongRecord class to hold song data.
- Holds all song information from the CSV.

Node

- Node class for the doubly linked list node.
- Attributes: data (a SongRecord object), prev, next.

DoublyLinkedList

DoublyLinkedList class to manage the playlist. Implements all playlist operations using nodes. Must include:

- `insert_front()`
- `delete_song(title)`

- `edit_song(title)`
- `rate_song(title, rating)`
- `play_songs()`
- `load_from_csv()`
- `store_to_csv()`
- `display()`

Main Menu Commands

The main menu should offer the following commands:

1. Load playlist from file
2. Store playlist to file
3. Display playlist
4. Insert new song
5. Delete a song
6. Edit a song
7. Sort playlist
8. Rate a song
9. Play songs
10. Shuffle play (extra)
11. Exit

After each command is completed, the menu must display again.

Playlist File Format

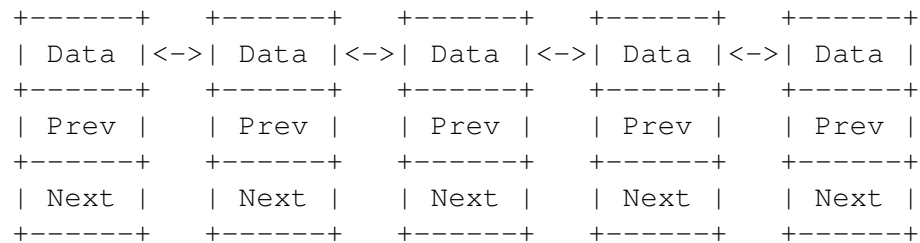
Your playlist must be read from a CSV file named `musicPlaylist.csv`. Each record contains:

- Artist (string)
- Album title (string)
- Song title (string)
- Genre (string)

- Duration (MM:SS format)
- Number of times played (int)
- Rating (int from 1 to 5)

III. Logical Block Diagram

Each node points to a song record and has links to previous and next nodes.



IV. Submission

Submit a single zipped Python project to the course submission system containing:

- Python source files (.py)
- A sample `musicPlaylist.csv` file

V. Grading Rubric (100 points)

- 5 pts – Proper class definitions and documentation
- 10 pts – Correct implementation of the linked list structure
- 5 pts – Looping back to the menu after each command
- 10 pts – Correct file loading
- 10 pts – Correct file saving
- 10 pts – Correct insertion of songs
- 10 pts – Correct deletion of songs
- 10 pts – Correct editing of song details
- 10 pts – Correct rating functionality
- 10 pts – Play songs in order

- 5 pts – Proper display of all records
- 5 pts – Sorting songs (optional: by title, artist, rating)